



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SHRI AMARAVATHI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

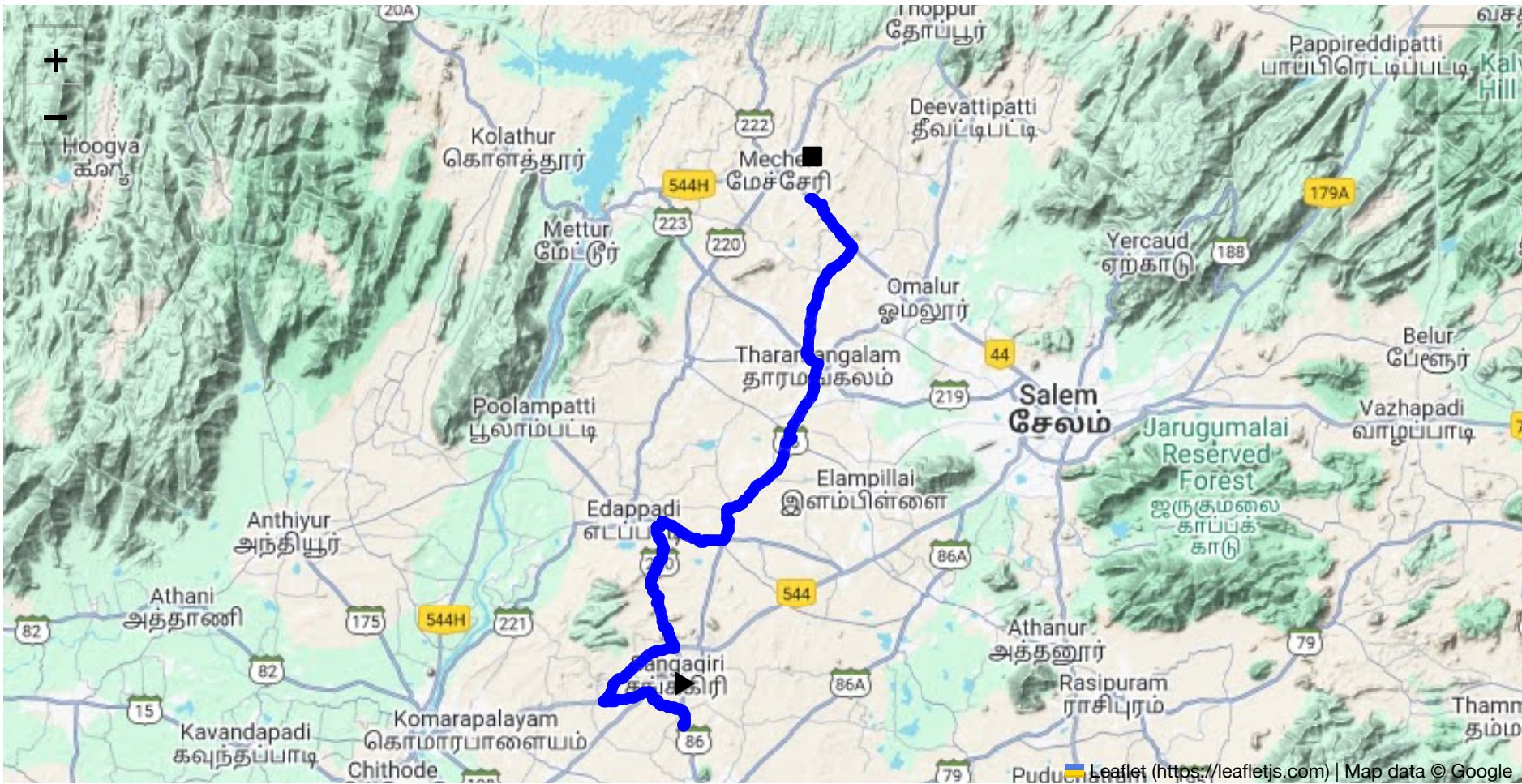
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 65.03 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours

Start: (11.4381, 77.8734)
End: (11.812703, 77.966036)



Welcome to the Journey Risk Management Study

Certainly! Below is a detailed analysis of the given route to help inform and train a truck driver transporting hazardous materials:

- Overview of the Route Map:**
 - The route travels from Sangagiri to Salem via Padaiveedu and Kurumbapatti. This route spans approximately 65.03 kilometers through rural and semi-urban areas with a mix of state highways and local roads.
- Typical Weather Conditions and Potential Weather-Related Hazards:**
 - Weather Conditions:** The region generally experiences tropical weather with hot summers, moderate winters, and a monsoonal climate.
 - Potential Hazards:** During the monsoon season (June to September), heavy rains may lead to slippery roads and potential flooding, particularly near low-lying areas around water bodies.
- Analysis of Traffic Patterns:**
 - Peak Hours:** Traffic congestion typically occurs during morning hours (8:00-10:00 AM) and evening hours (5:00-8:00 PM) when locals commute for work.
 - Congestion-Prone Areas:** Markets and junctions in towns along Bhavani Main Road and Idappadi Rd can experience significant congestion.
- Assessment of Road Quality and Infrastructure:**
 - The main roads are mostly paved but may have sections with potholes and uneven surfaces, particularly after heavy rainfall. Local roads towards rural areas may be narrower, requiring careful navigation for heavy vehicles.
- Suggestions for Alternative Routes for Emergencies:**
 - Consider utilizing NH44 as an alternative route if sections of the primary route become impassable. Ensure access to regional maps for additional smaller road alternatives.

6. **Summary of Local Regulations Affecting Hazardous Material Transport:**

- Compliance with local regulations includes obtaining necessary permits, displaying hazardous material placards, and adhering to speed limits specifically mandated for heavy vehicles. Certain roadways may have additional restrictions based on the nature of the transported materials.

7. **Overview of Historical Incidents:**

- The route has seen incidents related to heavy vehicles, often caused by overturning due to narrow roads or inadequate turns. Awareness of such historically problematic areas is important to ensure safety.

8. **Environmental Considerations and Sensitive Areas:**

- There may be sensitive ecological zones near Kurumbapatti and similar areas. It is crucial to minimize pollution and risks of spillage, particularly near water sources and protected lands.

9. **Analysis of Communication Coverage:**

- The primary towns and highway sections should generally provide stable communication networks. However, remote segments along rural areas may experience brief dead zones. Drivers should prepare by informing relevant contacts well ahead of these areas.

10. **Estimated Emergency Response Times:**

- Within town areas, emergency response times can range from 15 to 30 minutes. For remote segments, response times may extend to 45 minutes or more, depending on accessibility and distance from nearest emergency facilities.

11. **Overall Summary of Risk Assessment:**

- The route is moderately challenging due to varied road conditions, potential weather complications, traffic in towns, and the need for heightened awareness of environmental and regulatory requirements. Preparations should involve thorough route planning, continuous weather and traffic updates, and contingency plans for alternative routing and communication in less accessible areas. Transporters should ensure all regulatory compliances and safety measures are well understood and adhered to minimize potential risks during transport.

This detailed analysis should assist in thoroughly preparing for safe transport along the specified route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.57045, 77.90222	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	Medium	11.45358, 77.85781	30 KM/Hr
7	Turn	High	11.45353, 77.85697	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	Medium	11.45853, 77.85139	30 KM/Hr
9	Turn	Medium	11.45906, 77.85132	30 KM/Hr
10	Turn	Medium	11.46307, 77.84941	30 KM/Hr
11	Turn	Medium	11.46316, 77.84921	30 KM/Hr
12	Turn	High	11.45542, 77.81725	15 KM/Hr
13	Turn	High	11.45636, 77.81678	15 KM/Hr
14	Blind Spot	Blind Spot	11.49397, 77.86740	10 KM/Hr
15	Turn	Medium	11.50885, 77.86037	30 KM/Hr
16	Turn	Medium	11.52571, 77.85522	30 KM/Hr
17	Turn	Medium	11.52744, 77.85582	30 KM/Hr
18	Turn	Medium	11.55049, 77.85513	30 KM/Hr
19	Turn	Medium	11.55460, 77.85624	30 KM/Hr
20	Turn	High	11.57893, 77.85430	15 KM/Hr
21	Turn	Medium	11.57903, 77.85441	30 KM/Hr
22	Turn	Medium	11.58304, 77.90656	30 KM/Hr
23	Turn	Medium	11.64217, 77.95200	30 KM/Hr
24	Blind Spot	Blind Spot	11.69590, 77.97114	10 KM/Hr
25	Turn	High	11.70133, 77.96363	15 KM/Hr
26	Turn	Medium	11.73415, 77.96718	30 KM/Hr
27	Turn	High	11.77772, 77.99628	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
3	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
4	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
5	hospital	Veda Maruthuvamanai Hospital	11.6956155, 77.9678122	30 km/h	Medium
6	clinic	E.V.P Vairavel Clinic	11.6954975, 77.9667903	30 km/h	Medium
7	clinic	AMS Clinic	11.752664, 77.974974	30 km/h	Medium
8	hospital	Tholasampatti, Government Hospital	11.75638, 77.973813	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57045, 77.90222



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45358, 77.85781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49397, 77.86740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50885, 77.86037



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52571, 77.85522



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52744, 77.85582



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55049, 77.85513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55460, 77.85624



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58304, 77.90656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64217, 77.95200



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.69590, 77.97114



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70133, 77.96363



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73415, 77.96718



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77772, 77.99628

